



Enter cobots

Cobots, or collaborative robots, are robots that work alongside humans. They are generally lightweight, affordable, and easy to program. Most of them are just arms and wheels and look nothing like the Jetsons' maid Rosey—but they are quite endearing in their own way, and human workers welcome their assistance.

In the mid-90s, General Motors (GM) faced a challenge from the Occupational Safety and Health Administration to find ways to improve ergonomics in factories. Ergonomics was a pressing issue for most heavy manufacturing industries, as these problems ended up as injuries in the long run, hitting both workers' health and productivity.

Prasad Akella, who led the effort to develop cobots at GM, explained the problem in a media interview of that time: "If you tried doing something as simple as picking up and installing a 40-pound car battery, one a minute for eight hours a day, 200 days a year, your back is going to start feeling very sore." So, GM started working with researchers at Northwestern University and the University of California, Berkeley, to develop safe robots that could work with people, helping them with lifting heavy objects and other unergonomic tasks.

Michael Peshkin and J. Edward Colgate at Northwestern University ended up developing collaborative robots, and the team at University of California, Berkeley developed extenders or human power amplifiers. Colgate and Peshkin first called their people-friendly robots as programmable constraint machines (what an unfriendly name!) and later adopted the term 'cobot,' which was suggested by Brent Gillespie in a lab competition to come up with a better name.

It is clear from the story of its birth, that cobots were never intended to replace humans. They were only intended to work with humans to improve their efficiency and effectiveness by extending their strength

Potential applications

- Quality inspection
- Material handling
- Machine tending
- Palletising
- Dispensing
- Warehouse automation
- Kitchen automation
- Greenhouse automation
- Automated kiosks
- Agricultural processes
- Cleaning and sanitisation
- Lab automation
- Healthcare assistance

and precision and augmenting their intelligence with data.

"The idea is to have a world where people work with robots, not work like robots. Unlike industrial robots that require safety fences, cobots are capable of working alongside human workers without the need for the installation of safety cages (upon risk assessment). Cobots can take on dull, dirty, and dangerous tasks, relieving human workers to focus on higher-value tasks. They can, for example, take on repetitive tasks like packing and palletising, or delicate and risky jobs such as handling grippers, spot welding, screwing, and other difficult activities like lifting heavy parts," says Sougandh K.M., Country Manager - India, Universal Robots (UR).

Workers who do the same task requiring a similar physical movement constantly often suffer from what is known as repetitive strain injury (RSI). Cobots are ideal for such jobs. Advanced vision systems and precision tools enable cobots to do high-precision jobs that can be very stressful and time-consuming for humans.

Venkatesh Gurappa, Founder Director, Bangalore Robotics Pvt Ltd, India and Bangalore Robotics Science and Research gUG, Grünwald, Germany, and President, BlrAUV Inc, USA, explains further that cobots are small, cheaper than traditional robots, efficient, intelligent and fast. "They will be very helpful in industrial production units, where they can assist the technicians very efficiently and reliably. They can see and reach dif-

ficult areas easily. A cobot's mechanical arm can have more movable joints than the human hand, and so can reach inside the depths for a screw or bolt hidden behind many parts. Besides, the cobot can be trained to do any particular sequence of tasks. It can learn and mimic the actions of the human master," he says.

Guruprasad S., Associate Vice President, ABB India Limited, adds, "Cobots are designed for low payload applications such as handling small parts and inspection tasks. But there is so much more: AI integration and use of software can transform any connected industrial robot with higher payloads and faster speeds into a collaborative one. From testing samples in labs to making computer chips, collaborative robots (cobots) are working together with humans to transform our lives. Cobots are taking on repetitive and mundane tasks and helping create the flexibility and scalability for small and medium-sized enterprises to grow."

What makes cobots special

A cobot is a robot that meets certain requirements. First, of course, is its ability to work along with humans without caging—and this is no easy feat as humans are known for their unpredictability. And then, there is everything else that makes it a worker's friend and not a threat.

Katherine Scott, Developer Advocate, Open Robotics, explains, "First and foremost, cobots need to be safe to operate alongside their human colleagues. The primary implication of this is that often cobots aren't as fast or as strong as industrial robots to avoid potentially dangerous situations. Cobots usually have more sophisticated control systems that allow them to interact with humans safely: A good cobot will stop or slow down if it bumps into another object. Since cobots can work alongside people, they are capable of degrees of autonomy that weren't available to industrial robots. An autonomous mobile robot, for example, can be tasked with mov-